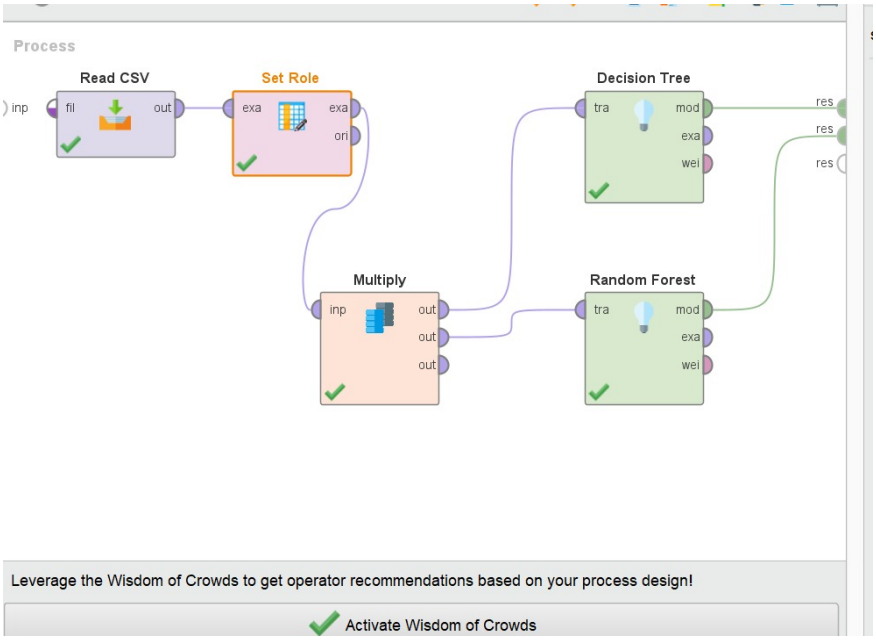
Shape Prediction using Decision Tree and Random Forest in Altair AI Studio

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Task

The data has **15 rows** with the columns **shape**, **color**, **outline** and **dot**, where the shape is known. The class work has two parts: **Part 1** builds a Decision Tree and a Random Forest from the 15 rows, and **Part 2** uses the Random Forest to predict the shape of 2 more rows whose shape is missing (?).

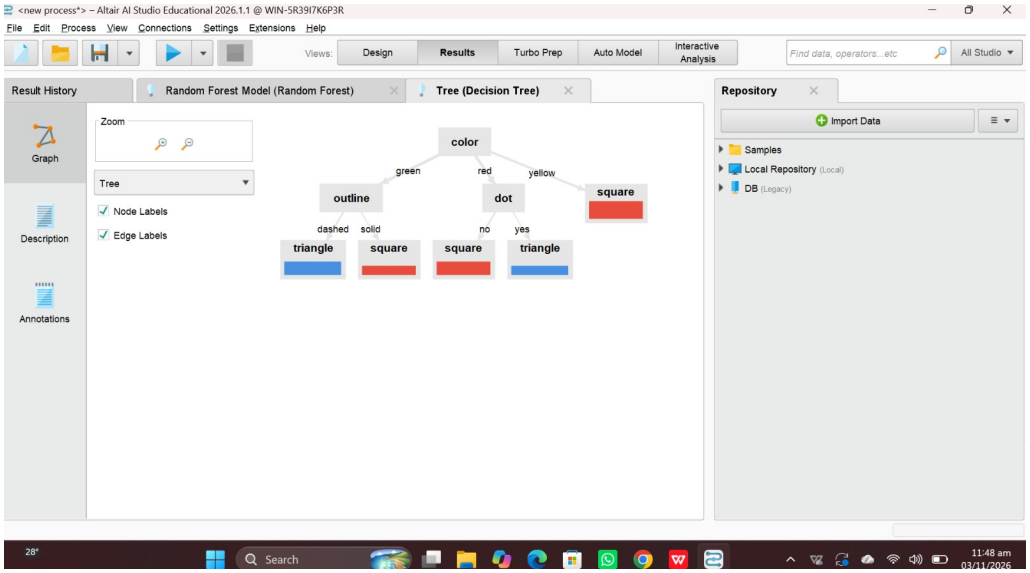
Part 1: Decision Tree and Random Forest



Process design for Part 1.

1. **Read CSV:** loads the 15 rows.
2. **Set Role:** sets **shape** as the label, the column to predict. Color, outline and dot stay as regular inputs.
3. **Multiply:** makes two copies of the data, one for each model.
4. **Decision Tree:** builds a single tree from the data.
5. **Random Forest:** builds many trees from the same data and combines their results.

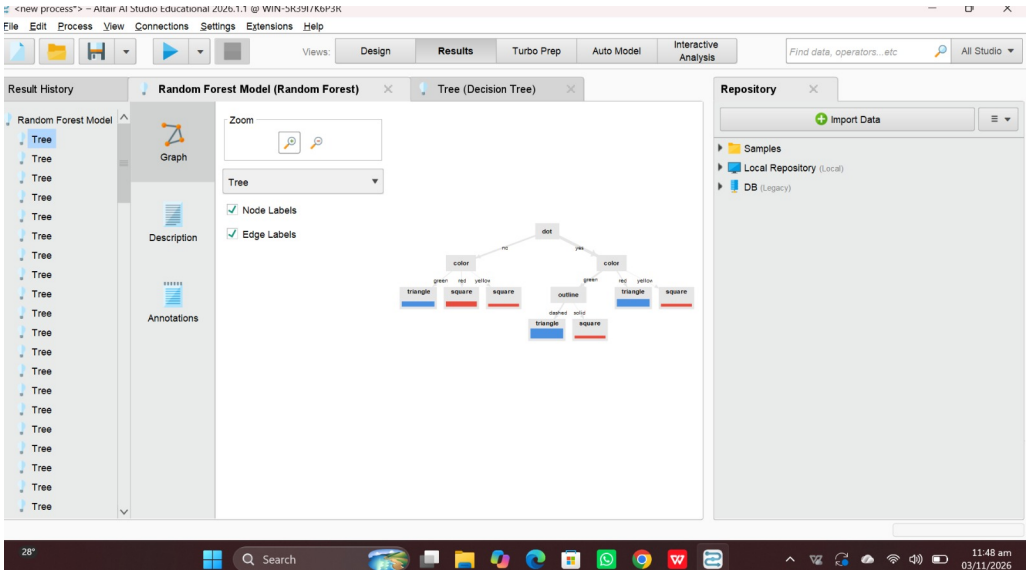
Decision Tree Result



Decision Tree (Graph view).

The tree first asks the **color**. If it is **yellow**, the shape is a square. If it is **green**, it checks the **outline** (dashed = triangle, solid = square). If it is **red**, it checks the **dot** (no = square, yes = triangle).

Random Forest Result

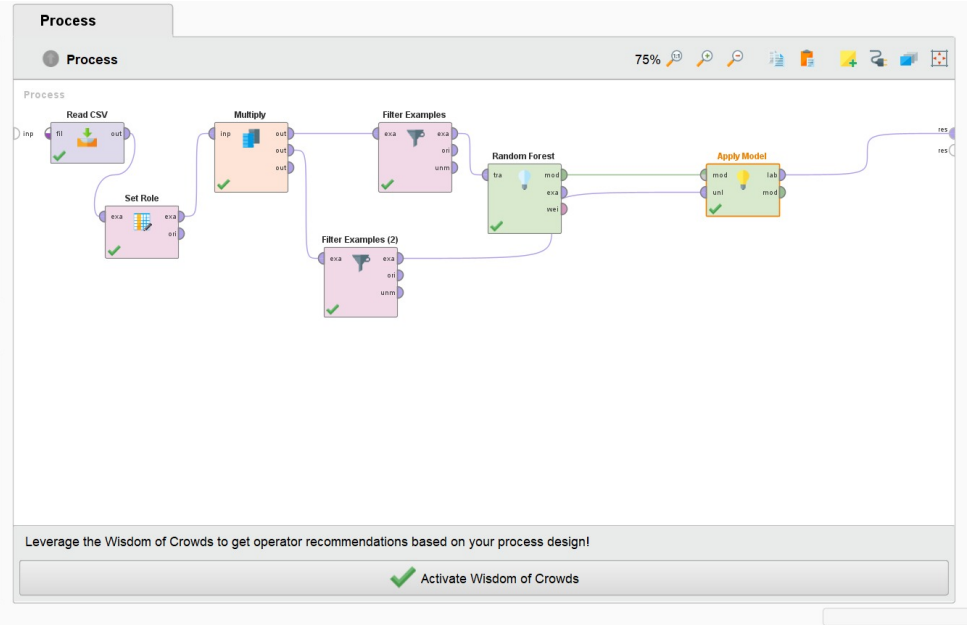


Random Forest model: the list on the left shows the individual trees, and the graph shows one selected tree.

The Random Forest builds many trees. Each tree uses a random part of the data, and the final answer is the combined vote of all trees. The tree shown starts with **dot** and then uses **color** and **outline**.

Part 2: Predicting the Unknown Shapes

Two more rows have color, outline and dot, but their **shape is missing (?)**. The Random Forest trained on the 15 known rows is used to **predict the shape** of these 2 rows.



Process design for Part 2.

1. Read CSV: loads the data file.
2. Set Role: sets **shape** as the label. Color, outline and dot stay as regular inputs.
3. Multiply: makes two copies of the data, one for learning and one for prediction.
4. Filter Examples (top): keeps only the rows where the shape is known (the 15 rows) as training data.
5. Random Forest: learns from these rows by building many decision trees.
6. Filter Examples (2): keeps only the 2 rows where the shape is missing.
7. Apply Model: applies the trained Random Forest model (mod) to those 2 rows (unl) to predict their shape.

Result

ExampleSet (Apply Model)

Open in Turbo Prep Auto Model Interactive Analysis

Row No.	shape	prediction(s...	confidence(...	confidence(...	color	outline	dot
1	?	square	0.077	0.923	yellow	solid	yes
2	?	triangle	0.686	0.314	red	solid	yes

Apply Model output: predicted shape for the 2 unknown rows.

The shape column still shows **?** because it was unknown. The **prediction** column gives the answer, and the **confidence** columns show how sure the model is for each shape.

• Row 1 (yellow, solid, dot): predicted **square** with confidence 0.923.
• Row 2 (red, solid, dot): predicted **triangle** with confidence 0.686.

Conclusion: using a Random Forest trained on the 15 known rows, the first unknown shape was predicted to be a **square** and the second a **triangle**.